

MATH UN1101.002 – Calculus I

SPRING 2026

Days & Times	Section 002
Room	MW 1:10-2:25PM
Class Dates	MACY HALL 263, TEACHER'S COLLEGE
	JAN 20, 2026 – MAY 15, 2026

Instructor: DUSA McDUFF

Email: dusa@math.columbia.edu

Office Hours: 2:40-4 on Mon and 10:30-12 on Wed in Math Bldg, Room 612, and Friday 4-5 on Zoom ([link in Courseworks](#)) or by appointment

Teaching Assistants:

	Email	Office Hours
Vivian Yu	qy2266	Tues 4-6 pm, Frid 9-10 am, 502 Milstein
Lynn Kim	lhk2133	Wed 7-8pm, Thurs 12- 1 pm, 502 Milstein

Course Webpage: CourseWorks - [MATH UN1101.002](#)

Course information including announcements, assessment due dates, and term grades will be available on the course webpage. Supplementary files are posted in the FILES tab. **You are expected to check it regularly.**

Course Description

Functions, limits, derivatives, introduction to integrals. An understanding of pre-calculus will be assumed.

Prerequisites: You are strongly advised to review basic concepts from preCalculus at (or before) the beginning of the semester: for example, you might work through (some of) the free Calculus [BootCamp](#) on the Columbia Mathematics department website. There will also be an in-person bootcamp during the first few days of the semester (details to follow).

Course Objectives

To develop a conceptual understanding of the fundamental principles of differential and integral calculus and explore their relevance to other branches of natural sciences.

Topics

The sequence of topics covered is as follows.

- (i) Functions. Exponential functions. Inverse functions and logarithms.
- (ii) Limits and continuity of functions of one variable.
- (iii) Differentiation of functions of one variable.
- (iv) Applications of differentiation.
- (v) Integrals. The Fundamental Theorem of Calculus. Applications.

A detailed class schedule is posted separately. (This includes the due dates for the assignments.)

Course Goals

Specific learning outcomes will be provided for each topic. One of the main objectives of this course is to foster the development of:

- A conceptual understanding of limits, derivatives, and integrals; and the ability to use logical evidence to construct proofs and solve problems involving these concepts.
- The ability to recognize and apply the underlying notions and principles of calculus in understanding changing quantities in mathematics and other natural sciences.
- The ability to think critically and creatively about complex problems.
- The ability to read mathematics with understanding and use mathematical terminology to communicate mathematical ideas with clarity both orally and in writing.

Textbook

REQUIRED:

- *WebAssign for Stewart's Calculus: Early Transcendentals, 9th edition*

OPTIONAL:

- James Stewart's *Calculus: Early Transcendentals, 9th edition* textbook (printed copy)
- *Student Solutions Manual for Stewart's Calculus: Early Transcendentals.*

WebAssign

You are **required** to have a WebAssign account in order to complete the homework problem sets.

The course code for this section is columbia 1326 4085

WebAssign access includes access to the electronic version for the textbook. For optimal value, the Mathematics Department recommends purchasing **Cengage Unlimited**, which for Columbia students includes multi-term WebAssign usage and grants access to all texts from the publisher (some of which may be used in other courses).

Course Delivery and Support

Lectures

The lectures, class handouts, and WebAssign homeworks represent the essential content of the course and, unless otherwise indicated, you are responsible for **all the material** covered.

Office Hours

The instructor will hold weekly office hours some in person, some on Zoom, hours are listed above. For info on Zoom OH see the listings in Courseworks.

The TAs will hold weekly hours in the Math Help Room in room 502 Milstein Center, Salmasi Computational Science/Math Collaboration Space (Barnard Campus). You are encouraged to come to their hours with any questions you may have.

Check the [Help Room](#) for Calculus I office hours. Also check [Barnard Help Room](#) for the Barnard tutoring hours (which are open to all Calculus students).

Course Assessment

Homework

Homework problem sets, assigned after most lectures (and starting TBA), will be available on [WebAssign](#). Each problem set covers about one section in the textbook and requires about two hours to complete. Each homework question will allow for 5 attempts. Each problem set can be extended (without penalty) within a day of the deadline for one additional day. After the due date (or extension), you will still be able to complete the questions or try another version, but you will not receive credit for it, unless explicit permission has been granted by the instructor for a further extension due to some compelling reason. *Homework counts toward 17% of your final grade.*

In-class work

To encourage attendance and class participation, a worksheet will be handed out in class and there will be time in class to try some of the problems. Every week or so you will also be asked to hand in a sheet with comments and questions: this will be used to take attendance. If you cannot attend class because of illness (or an away game etc) please email me (dusa@math.columbia.edu) before class. *Attendance will count toward 3% of your final grade.*

Assignments

There will be five written assignments throughout the term, one for each module: A1 (functions), A2 (limits and continuity), A3 (differentiability), A4 (applications of differentiation), A5 (integrability). Collaboration and discussion with your classmates is encouraged, but **you must write up assignments individually and in your own words**. Completed assignments should be uploaded to Gradescope. Late assignments will not be accepted, except for special circumstances and with documentation from a dean or university official. *Assignments count toward 15% of your final grade.* For the timing of these assignments see the detailed course Schedule.

Additional Practice

Additional **practice problems** will be made available throughout the term and there will also be sets of **recommended problems** before each test and the final exam. The practice problems and the recommended problems are **not** to be submitted for grading.

Extra credit

There will be opportunities to earn roughly **5% extra credit** toward your final grade. Examples of such opportunities include: 1) an optional WebAssign problem set, or 2) completing an applied project or 3) doing extra problems on the assignments.

Tests and Final Exam

These are in person, written exams, with no calculators, cheat sheets or collaboration allowed.

There will be **two midterm tests** and **one cumulative final exam**.

- Each midterm will be a **75-minute exam**. Midterms are **(tentatively) scheduled** on Wed. Feb 18 and Wed. Apr. 1. during normal class hours.
- The final exam will be a **2 hour 50 min exam** and is **provisionally scheduled** by the University on TBA *Writing the final exam is mandatory.*

Students will occasionally have two final exams scheduled for the same time. If one of the exams is a departmental exam for a class with several sections, the student should contact the department to arrange a make-up exam. Similarly, no student is obliged to take three exams on any given calendar day. Students with three exams on the same day should print their exam schedule from Vergil and present it to each instructor or department. An attempt will then be made to arrange for one of the instructors to schedule a make-up exam on a different day. **Examinations are not rescheduled to accommodate travel plans..**

Grading

Evaluation and Grading

The formula of the course grade (excluding the extra credit) is the highest of the following:

Grading Scheme A		Grading Scheme B		Grading Scheme C	
WebAssign	17%	WebAssign	17%	No WebAssign	0%
Attendance	3%	Attendance	3%	No In-class work	0%
Assignments	15%	Assignments	15%	Assignments	15%
Midterms (2×15% each)	30%	Midterm (best of two)	15%	Midterms (2×20% each)	40%
Final Exam	35%	Final Exam	50%	Final Exam	45%
Total	100%	Total	100%	Total	100%

The letter grade for the WebAssign, In-class work and Assignments parts of the class course will be assigned based on the following ranges:

Letter Grade	Range
A+	98% – 100%
A	94% – 97.9%
A-	90% – 93.9%
B+	87% – 89.9%
B	84% – 86.9%
B-	80% – 83.9%
C+	77% – 79.9%
C	74% – 76.9%
C-	70% – 73.9%
D	60% – 69.9%
F	0% – 59.9%

Similar rubrics will be made up and publicized after each Midterm and the Final, and the cut offs for the final grade will be computed accordingly for each of the three grading rubrics.

The instructor and university reserve the right to change the weight of any portion of the final grade scheme for the course, under special circumstances. *Any modification that becomes necessary will have to apply to all students equally. Reasonable notice for any modification will be given with explanation and the opportunity to comment on the changes.*

Please note! Once the final exam is written, the final grade cannot be adjusted to take into account any special situation.

Detailed class schedule

This is posted in a separate file.

School and University Policies

Message for students

Columbia University is committed to an **inclusive and respectful community**. These principles and expectations extend to out of classroom and online activities, including electronic chat groups, video calls and other learning platforms.

If you are concerned about your classroom or out of classroom experiences, the [Office of Equal Opportunity and Affirmative Action \(EOAA\)](#) is available to advise and assist students who may be experiencing any equity, accessibility, inclusion, harassment, discrimination or sexual violence concerns. To [make an appointment](#), or [file a report](#), or find more information for students, visit the [Student Conduct and Community Standards](#) webpage.

Student Attendance

Students are expected to attend every session of the classes, and to keep up with the syllabus. They are required to attend the Midterm exams. Late homework will not be accepted, except under very special circumstances. Students who must miss a class due to religious observance, illness, or family emergency, should notify the instructor, in advance if at all possible, so that appropriate arrangements can be made

Academic Integrity

A Columbia University education has two complementary elements: the intellectual development of bodies of knowledge and habits of mind, and the overall development of moral character and personal ethics. These elements are supported by a commitment to academic integrity.

Academic integrity defines an intellectual community and its educational mission. As members of the intellectual community of Columbia University, students are expected to respect and acknowledge the intellectual work of others and denote clearly how their own work relates to that of others.

Violating the standards of academic integrity severely inhibits a student's opportunity to mature academically, professionally, and socially. Consequently, a violation of academic integrity is one of the most serious offenses a student can commit at the University.

It is your responsibility to understand what constitutes academic dishonesty and how disciplinary processes for violations of academic integrity are handled. For more details visit:

Barnard College: [Students' Roles and Responsibilities](#) and [Policy on Academic Integrity in Online Courses](#)

Columbia College: [Academic Integrity](#) and [Columbia College - Columbia Engineering Academic Integrity](#)

Columbia School of General Studies: [Academic Integrity and Community Standards](#)

Use of Artificial Intelligence (AI)

Students are not allowed to use AI generative or machine learning tools to complete deliverables for this course. In accordance with Columbia University's academic integrity policy, individuals must complete their own work and properly acknowledge the circumstances, ideas, sources, and assistance upon which that work is based.

Online Elements

In this course we will be using *email* and online platforms such as *CourseWorks*, *Gradescope*, and *Zoom*. Students should be aware that, when they access the electronic components of this course, private information such as their first and last names, UNI, and program affiliation may become apparent to other students in the same course.

Continuation in this course will be deemed consent to this disclosure. If you have any questions or concerns about such disclosure please discuss them with the course instructor.

Class Recordings

If any classes are given by Zoom, they will be recorded, unless this is deemed to be obstructive to the learning process. If the recording is posted, it is confidential and it is prohibited to share the recording outside of the class. Zoom office hours are NOT recorded.

Diversity Statement

It is our intent that students from all diverse backgrounds and perspectives be well-served by this course, that students' learning needs be addressed both in and out of class, and that the diversity that the students bring to this class be viewed as a resource, strength and benefit. It is our intent to present materials and activities that are respectful of diversity: gender identity, sexuality, disability, age, socioeconomic status, ethnicity, race, nationality, religion, and culture.

Academic Accommodation of Students with Disabilities

[Disability Services](#) facilitates equal access for students with disabilities by coordinating accommodations and support services, and cultivating a campus culture that is sensitive and responsive to the needs of students.

In order to receive disability-related academic accommodations, CC and GS students must first be registered with [Disability Services \(DS\)](#). More information on the DS registration process is available online at <https://health.columbia.edu/ods>.

Faculty must be notified of registered students' accommodations before exams or other accommodations will be provided. Students who have, or think they may have, a disability are invited to contact Disability Services for a confidential discussion at (212) 854-2388 (Voice/TTY) or by email at disability@columbia.edu. Barnard students are invited to visit the [Center for Accessibility Resources & Disability Services \(CARDS\)](#).

In order to receive disability-related academic accommodations for this course, students must first be registered with their school Disability Services (DS) office. Detailed information is available online for both the [Columbia](#) and [Barnard](#) registration processes. Refer to the appropriate website for information regarding deadlines, disability documentation requirements, and [drop-in hours](#) (Columbia)/[intake session](#) (Barnard).

For this course, students registered with the Columbia DS office can refer to the "Courses that do not require professor signature" section of the DS [Testing Accommodations](#) page for more information about accessing their accommodations.

Academic Accommodation for Religious Observances

It is the policy of Columbia University to respect the religious beliefs of its faculty, students, and staff. In accordance with New York State law, no student may be penalized for absences due to religious beliefs.

Students are required to inform their instructors of their plans to observe a religious holiday **at the beginning of the semester**, so that instructor and student have plenty of time to plan for any necessary alternative arrangements. It is important for students to understand that, no matter what absences might be accommodated, they will still be responsible for all course requirements, which may include in-class participation or other assignments that will need to be made up in some alternative way.

Columbia University Information Technology

[Columbia University Information Technology](#) (CUIT) provides Columbia University students, faculty and staff with central computing and communications services. Students, faculty and staff may access [University-provided and discounted software downloads](#).

Columbia University Library

Columbia's extensive [library system](#) ranks in the top five academic libraries in the nation, with many of its services and resources available online.

Disclaimer

Extreme Circumstances

The instructor and university reserve the right to modify elements of the course during the term. The university may change the dates and deadlines for any or all courses in extreme circumstances. If either type of modification becomes necessary, reasonable notice and communication with the students will be given with explanation and the opportunity to comment on changes. It is the responsibility of the student to check their University email and the course website during the term and to note any changes.